2nd Midterm · Unit 6 · Heat and wave equations by separation · 5 points · 45 min · Standard exam level

P12 — Heat equation with a constant source and fixed, different end temperatures: steady state, separation of variables for the deviation, series solution and the time constant of the slowest mode

A thin laterally insulated rod occupies $0 \le x \le L$. Its ends are held at the constant temperatures $u(0,t) = T_A$ and $u(L,t) = T_B$. For $t < 0$ the rod has been left long enough to reach the equilibrium with no internal heat source. At $t = 0$ an electric current is switched on that heats the rod uniformly, so that for $t > 0$ the temperature $u(x,t)$ satisfies

$$u_t = k\,u_{xx} + Q, \qquad 0 < x < L,\ t > 0,$$

with constant diffusivity $k > 0$ and constant source $Q > 0$ (in K/s).

(a) Write the initial temperature $u(x,0)$. Find the steady state $u_s(x)$ of the problem for $t > 0$ and the steady temperature at the midpoint.

(b) Write the problem satisfied by $v = u - u_s$. Separate variables in it, $v = X(x)T(t)$, and discuss the three cases of the sign of the separation constant to find all eigenvalues and eigenfunctions.

(c) Obtain $u(x,t)$ as a series with all coefficients in closed form.

(d) Find $\lim_{t\to\infty} u$, the time constant τ_1 of the slowest mode, and, keeping only that mode, the time t_{99} after which the midpoint has completed 99% of its final rise. Numbers: $L = 0.5$ m, $k = 1.0 \times 10^{-4}$ m^2/s, $Q = 0.02$ K/s, $T_A = 20$ °C, $T_B = 40$ °C.

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknown $u(x,t)$, independent variables $x \in [0,L]$, $t \ge 0$. One-dimensional conduction, constant properties, uniform volumetric source.
- Operator: linear, second order, parabolic; **non-homogeneous** twice: the source Q in the equation and the Dirichlet data T_A, T_B (non-zero). Separation of variables needs homogeneous boundary conditions, so the inhomogeneities are removed first with the steady state.
- Data: L [m], k [m^2/s], Q [K/s], T_A, T_B [°C or K].

Phase 2. Canonical formulation

$$u_t = k u_{xx} + Q, \quad u(0,t) = T_A, \quad u(L,t) = T_B, \quad u(x,0) = u_0(x)$$

Split $u(x,t) = u_s(x) + v(x,t)$, where u_s absorbs the source and the boundary data and v solves a homogeneous problem with homogeneous Dirichlet conditions (Haberman Sec. 7.2).

Part (a) · 1 point

Initial temperature and steady state

Step 1. Initial profile

Before the source is switched on the rod is in equilibrium, $u_t = 0$, so $k u'' = 0$ and $u = \alpha + \beta x$. The end values give $\alpha = T_A$, $\alpha + \beta L = T_B$, hence $u_0(x) = T_A + (T_B - T_A)x/L$ (Haberman Sec. 1.4).

Step 2. Steady state with the source

$u_s'' = -\dfrac{Q}{k}$. Integrating once: $u_s' = -\dfrac{Q}{k}x + C_1$; again: $u_s = -\dfrac{Q}{2k}x^2 + C_1 x + C_2$.

Step 3. Constants

$u_s(0) = C_2 = T_A$. $u_s(L) = -\dfrac{QL^2}{2k} + C_1 L + T_A = T_B$, so $C_1 = \dfrac{T_B - T_A}{L} + \dfrac{QL}{2k}$. Substituting and grouping $\dfrac{Q}{2k}(Lx - x^2)$:

$$u_s(x) = T_A + (T_B - T_A)\frac{x}{L} + \frac{Q}{2k}\,x(L - x)$$

Check: $k u_s'' = k \cdot \dfrac{Q}{2k} \cdot (-2) = -Q$, and $x(L-x)$ vanishes at both ends.

Step 4. Midpoint

$u_s(L/2) = T_A + \dfrac{T_B - T_A}{2} + \dfrac{Q}{2k} \cdot \dfrac{L}{2} \cdot \dfrac{L}{2} = \dfrac{T_A + T_B}{2} + \dfrac{QL^2}{8k}$. Numbers: $\dfrac{0.02 \cdot 0.25}{8 \cdot 10^{-4}} = 6.25$ K, so $u_s(L/2) = 36.25$ °C.

KEY RESULTS

- $u_0(x) = T_A + (T_B - T_A)\dfrac{x}{L}$.
- $u_s(x) = T_A + (T_B - T_A)\dfrac{x}{L} + \dfrac{Q}{2k}\,x(L-x)$.
- $u_s(L/2) = \dfrac{T_A + T_B}{2} + \dfrac{QL^2}{8k}$; with the numbers $30 + 6.25 = 36.25$ °C.

Part (b) · 1.5 points

Problem for v and separation of variables

Step 1. Homogeneous problem

$v_t = u_t - 0 = ku_{xx} + Q - (ku_s'' + Q) = k(u - u_s)_{xx} = kv_{xx}$. At the ends $v(0,t) = T_A - T_A = 0$ and $v(L,t) = T_B - T_B = 0$. Initially $v(x,0) = u_0 - u_s = -\dfrac{Q}{2k}x(L-x)$: the linear parts cancel.

Step 2. Separation

$v = X(x)T(t)$ gives $XT' = kX''T$, and dividing by kXT:

$$\frac{T'}{kT} = \frac{X''}{X} = -\lambda$$

The left side depends only on t and the middle only on x, so both equal a constant. Hence $T' = -\lambda kT$ and $X'' + \lambda X = 0, X(0) = X(L) = 0$ (the conditions on v with $T \not\equiv 0$).

Step 3. Case lambda < 0

Write $\lambda = -\mu^2, \mu > 0$: $X = a\cosh \mu x + b\sinh \mu x$. $X(0) = a = 0$; $X(L) = b\sinh \mu L = 0$ with $\sinh \mu L > 0$, so $b = 0$. Only the trivial solution.

Step 4. Case lambda = 0

$X'' = 0, X = a + bx$. $X(0) = a = 0, X(L) = bL = 0 \Rightarrow b = 0$. Trivial.

Step 5. Case lambda > 0

$\lambda = \mu^2$: $X = a\cos \mu x + b\sin \mu x$. $X(0) = a = 0$; $X(L) = b\sin \mu L = 0$ with $b \neq 0$ requires $\mu L = n\pi$. Hence

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2, \qquad X_n(x) = \sin \frac{n\pi x}{L}, \qquad n = 1, 2, 3, \ldots$$

and $T' = -k\lambda_n T$ integrates to $T_n(t) = e^{-k\lambda_n t}$.

KEY RESULTS

- $v_t = kv_{xx}, v(0,t) = v(L,t) = 0, v(x,0) = u_0 - u_s = -\dfrac{Q}{2k}x(L-x)$.
- $\lambda < 0$ and $\lambda = 0$: only $X \equiv 0$.
- $\lambda_n = \left(\dfrac{n\pi}{L}\right)^2, X_n = \sin \dfrac{n\pi x}{L}, T_n = e^{-kn^2\pi^2 t/L^2}, n = 1, 2, \ldots$

Part (c) · 1.5 points

Series solution

Step 1. Superposition

Each product X_nT_n satisfies $v_t = kv_{xx}$ and the zero end conditions, and so does any convergent combination:

$$v(x,t) = \sum_{n=1}^{\infty} c_n \sin\frac{n\pi x}{L}\, e^{-kn^2\pi^2 t/L^2}$$

At $t=0$: $-\dfrac{Q}{2k}x(L-x) = \sum c_n \sin\dfrac{n\pi x}{L}$, so by orthogonality $c_n = \dfrac{2}{L}\displaystyle\int_0^L \Big(-\frac{Q}{2k}\Big)x(L-x)\sin\frac{n\pi x}{L}\,dx$.

Step 2. The integral

Write $\omega = n\pi/L$. With $u = x(L-x) = Lx - x^2$, $dv = \sin\omega x\,dx$ ($du = (L-2x)\,dx$, $v = -\cos\omega x/\omega$):

$$\int_0^L (Lx-x^2)\sin\omega x\,dx = \Big[-\frac{(Lx-x^2)\cos\omega x}{\omega}\Big]_0^L + \frac{1}{\omega}\int_0^L (L-2x)\cos\omega x\,dx$$

The bracket is 0 because $Lx-x^2$ vanishes at $x=0$ and $x=L$. Next, $u = L-2x$, $dv = \cos\omega x\,dx$ ($du = -2\,dx$, $v = \sin\omega x/\omega$):

$$\int_0^L (L-2x)\cos\omega x\,dx = \Big[\frac{(L-2x)\sin\omega x}{\omega}\Big]_0^L + \frac{2}{\omega}\int_0^L \sin\omega x\,dx = 0 + \frac{2}{\omega}\Big[-\frac{\cos\omega x}{\omega}\Big]_0^L = \frac{2}{\omega^2}\big(1-\cos n\pi\big)$$

since $\sin\omega L = \sin n\pi = 0$. Hence $\displaystyle\int_0^L x(L-x)\sin\omega x\,dx = \frac{2(1-(-1)^n)}{\omega^3} = \frac{2L^3(1-(-1)^n)}{n^3\pi^3}$ and

$$\frac{2}{L}\int_0^L x(L-x)\sin\frac{n\pi x}{L}\,dx = \frac{4L^2(1-(-1)^n)}{n^3\pi^3}$$

Step 3. Coefficients and solution

Multiplying by $-\dfrac{Q}{2k}$: $c_n = -\dfrac{2QL^2(1-(-1)^n)}{k\pi^3 n^3}$, which is $-\dfrac{4QL^2}{k\pi^3 n^3}$ for odd n and 0 for even n. Adding u_s :

$$u(x,t) = T_A + (T_B - T_A)\frac{x}{L} + \frac{Q}{2k}x(L-x) - \frac{4QL^2}{k\pi^3}\sum_{n=1,3,5,\dots}\frac{1}{n^3}\sin\frac{n\pi x}{L}\,e^{-kn^2\pi^2 t/L^2}$$

The coefficients decay as $1/n^3$ (the odd extension of $x(L-x)$ is continuous with a continuous derivative), so the series converges fast even at $t = 0$.

KEY RESULTS

- $\dfrac{2}{L}\displaystyle\int_0^L x(L-x)\sin\frac{n\pi x}{L}\,dx = \frac{4L^2}{n^3\pi^3}\bigl(1-(-1)^n\bigr)$: $\dfrac{8L^2}{n^3\pi^3}$ for odd n, 0 for even n.
- $c_n = -\dfrac{4QL^2}{k\pi^3 n^3}$ for odd n, $c_n = 0$ for even n.
- $u(x,t) = T_A + (T_B - T_A)\dfrac{x}{L} + \dfrac{Q}{2k}x(L-x) - \dfrac{4QL^2}{k\pi^3}\displaystyle\sum_{n\ \text{odd}}\frac{1}{n^3}\sin\frac{n\pi x}{L}\,e^{-kn^2\pi^2 t/L^2}$.

Part (d) · 1 point

Long-time limit, time constant and the 99 % time

Step 1. Limit

Every term carries $e^{-kn^2\pi^2 t/L^2} \to 0$ and $\sum 1/n^3$ converges, so $v \to 0$ and $u \to u_s$: the rod forgets its initial state.

Step 2. Slowest mode

The $n = 1$ term decays as e^{-t/τ_1} with $\tau_1 = \dfrac{1}{k\lambda_1} = \dfrac{L^2}{k\pi^2} = \dfrac{0.25}{10^{-4}\cdot 9.8696} = 253.3$ s. The next non-zero mode, $n = 3$, has $\tau_3 = \tau_1/9$.

Step 3. Midpoint rise

The total rise of the midpoint is $u_s(L/2) - u_0(L/2) = \dfrac{QL^2}{8k} = 6.25$ K. At $x = L/2$ the deficit is

$$u_s\Bigl(\frac{L}{2}\Bigr) - u\Bigl(\frac{L}{2},t\Bigr) = \frac{4QL^2}{k\pi^3}\sum_{m=0}^{\infty}\frac{(-1)^m}{(2m+1)^3}e^{-(2m+1)^2 t/\tau_1}$$

At $t = 0$ the sum is $\pi^3/32$ (bank item P11), giving $\dfrac{4QL^2}{k\pi^3}\cdot\dfrac{\pi^3}{32} = \dfrac{QL^2}{8k}$, as it must. Keeping $m = 0$, the fraction still missing is $\dfrac{32}{\pi^3}e^{-t/\tau_1}$.

Step 4. The 99 % time

$\dfrac{32}{\pi^3}e^{-t_{99}/\tau_1} = 0.01 \Rightarrow t_{99} = \tau_1 \ln\dfrac{3200}{\pi^3} = \tau_1\ln(103.2) = 4.637\,\tau_1 \approx 1174.5$ s. At that time the $m = 1$ term is smaller than the $m = 0$ term by the factor $\frac{1}{27}e^{-8\cdot 4.637} < 10^{-17}$: the one-mode answer is exact for practical purposes.

KEY RESULTS

- $u(x,t) \to u_s(x)$ as $t \to \infty$.
- $\tau_1 = \dfrac{L^2}{k\pi^2} = 253.3$ s; the $n = 3$ mode decays 9 times faster.
- Midpoint: $\dfrac{u_s(L/2) - u(L/2,t)}{QL^2/(8k)} \approx \dfrac{32}{\pi^3}e^{-t/\tau_1}$, so $t_{99} = \tau_1 \ln\dfrac{3200}{\pi^3} = 4.637\,\tau_1 \approx 1174.5$ s (about 19.6 min).

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** $Q \to 0$ gives $u_s = u_0$ and $v \equiv 0$: nothing happens, as expected. $t \to \infty$: $u \to u_s$, a parabola above the straight line between T_A and T_B; with these numbers its maximum is interior, at $x^* = \dfrac{L}{2} + \dfrac{k(T_B - T_A)}{QL} = 0.25 + 0.2 = 0.45$ m, since $u_s'(x^*) = 0$.
- **Checks:** dimensions: $\dfrac{Q}{k}x^2$ is $\dfrac{\mathrm{K\ s^{-1}}}{\mathrm{m^2 s^{-1}}}\mathrm{m^2} = \mathrm{K}$; $\tau_1 = L^2/k$ is s. Each mode satisfies $v_t = kv_{xx}$ and vanishes at both ends; the series at $t = 0$ reproduces u_0 (checked numerically in the script).
- **Common mistakes in the exam:** (1) separating variables directly in u, whose boundary conditions are not homogeneous; (2) taking $v(x,0) = u_0$ instead of $u_0 - u_s$; (3) skipping the cases $\lambda \le 0$; (4) giving the time constant as L^2/k (forgetting π^2) or using the decay rate of v^2.